On Monday we focused on quarks...
we saw:

$$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$$

3 families →

$$\begin{array}{ccc} \begin{pmatrix} e^- \\ \nu_e \end{pmatrix} & \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix} & \begin{pmatrix} \tau \\ \nu_\tau \end{pmatrix} \\ \text{1st} & \text{2nd} & \text{3rd} \end{array}$$

Today we explore leptons, specifically
WEAK INTERACTION.

$$\boxed{n \to p\, e^- \bar{\nu}_e} \leftarrow \beta^- \text{ decay}$$

$$^{A}_{Z}X \to {}^{A}_{Z+1}Y\, e^- \bar{\nu}_e$$

$\boxed{\nu}$

← postulated by Pauli '30

"I have done a terrible thing I postulated a particle that cannot be detected"

continuous energy spectrum of e^- emitted in β decays.

observed by Cowan & Reines 1956

$\nu_\mu \neq \nu_e$ 1962 (Lederman - Schwartz - Steinberger)

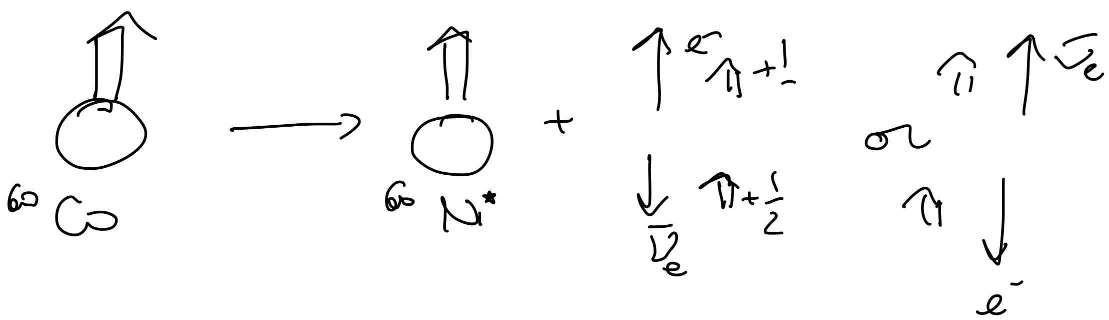
+ 1957 exp. Wu $\rightarrow$ P is violated in weak interaction.

+ 1957 exp Goldhaber $\rightarrow$ C is violated in weak interaction

or "the weak interaction acts differently on particles and their antiparticles"

⇒ C is not conserved in WEAK INTERACTION.

Let's look back at Wu's exp:



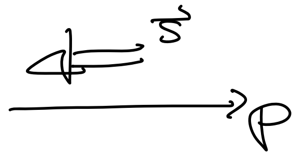
data compatible w/ this one

⇒ The intensity of the interaction varies if I mirror the final state ($\vec{x} \to -\vec{x}$)

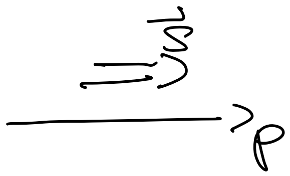
⇒ P is violated

Let's introduce now helicity:

$$h = \frac{\vec{p} \cdot \vec{S}}{|\vec{p}||\vec{S}|}$$ (handedness)

LH : 

$h = -1$

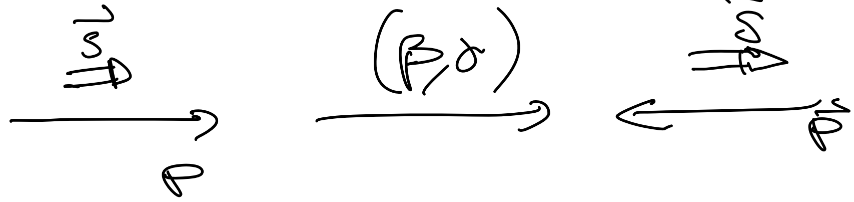
RH : 

$h = +1$

<u>NB</u> A priori h is not invariant b/c you can define a boosted R.F. where $\vec{p}$ is inverted (and hence h changes sign).

you can define this R.F. only for MASSIVE particles

$\Rightarrow$ if $m = 0$ helicity is a conserved quantity of WEAK INTERACTION.



Today we know that $m_\nu \sim 0$

hence h of ν_s is invariant

so if ν_s are produced with

defined h they won't change it

$\Rightarrow$ WEAK INTERACTION works

only w/ $h(\nu) = \begin{cases} +1 \\ -1 \end{cases}$

this is why we say that $\cancel{C}$

is violated!

GOLDHABER EXP. wants to measure

$h(\nu)$!

- which process I study? electron capture is a radioactive process

$$^{152}Eu + e^- \longrightarrow {}^{152}Sm^* + \nu_e$$

$$p + e^- \longrightarrow n + \nu_e$$

$+\frac{1}{2}\ \Uparrow\ \bigcirc\ {}^{152}Eu \longrightarrow \Downarrow -\frac{1}{2}\ \uparrow \nu_e \quad \Uparrow +1\ \downarrow Sm^*$ OR $\Uparrow \frac{1}{2}\ \uparrow Sm^* \quad \Downarrow -\frac{1}{2}\ \downarrow \nu_e$

$h(\nu) = -1 \qquad h(\nu_e) = +1$

$h(Sm^*) = -1 \qquad h(Sm^*) = +1$

$\Longrightarrow h(\nu_e) = h(Sm^*)$

If easier is equivalent to measure $h(Sm^*)$

· How does Sm^* decay?

$\tau(Sm^*) \sim 10^{-14}\,s \leftarrow$ ELM

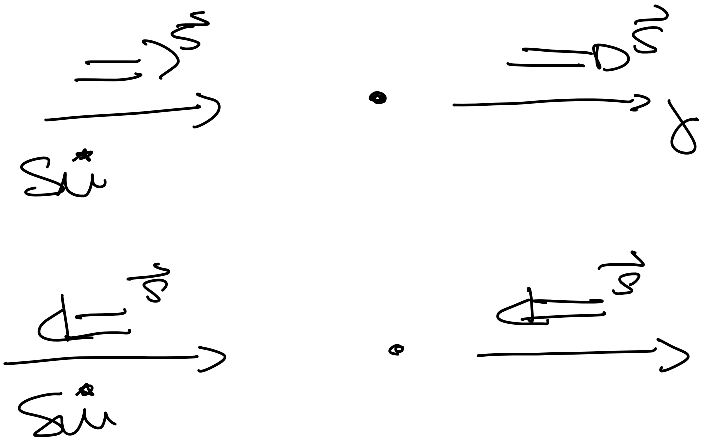
$^{152}Sm^* \longrightarrow {}^{152}Sm + \gamma$

SPIN: $+1 \longrightarrow 0 + 1$

$+1 \Uparrow \longrightarrow \bigcirc + \Uparrow$

$$h(Sn^*) = h(\gamma)$$

NB if γ emitted in the same flight direction of Sn^*

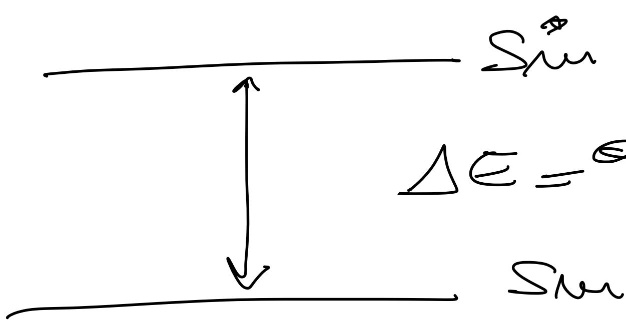


$h = 1$

$h = -1$

Basically $h(\gamma)$ is transfered to the $h(\gamma)$

• What is the $E(\gamma)$?



$\Delta E = 961$ keV

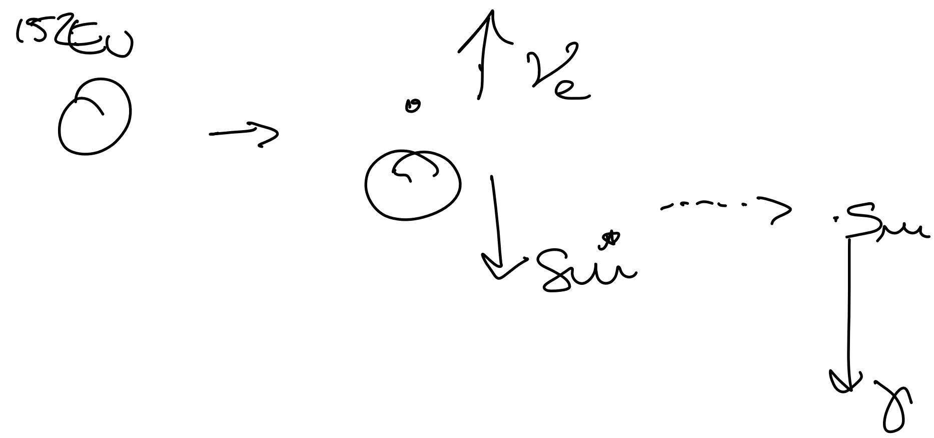
$\Gamma \sim 10^{-2}$ eV

However: in $Sm^* \to Sm + \gamma$ ← if at rest!
3.2 eV are lost from $E(\gamma)$ to give RECOIL of Sm.
Hence if γ encounter an Sm:

$$\gamma + Sm \cancel{\to} Sm^*$$

↑ cannot happen a priori. b/c. $E(\gamma) < 951$ keV

<u>But</u>: in the Eu decay, Sm^* is produced w/ a little boost in the $-z$ direction which allows to INCREASE the $E(\gamma)$ (if emitted in $-z$) enough to recover the 951 keV threshold!



$\Rightarrow$ <u>if</u> γ emitted in $-z$ it gets a

BOOST

$\downarrow$

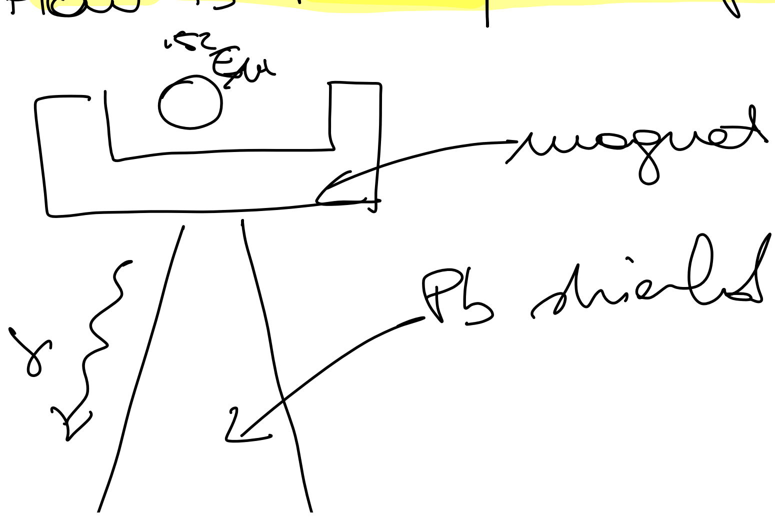
$E(\gamma) \nearrow$

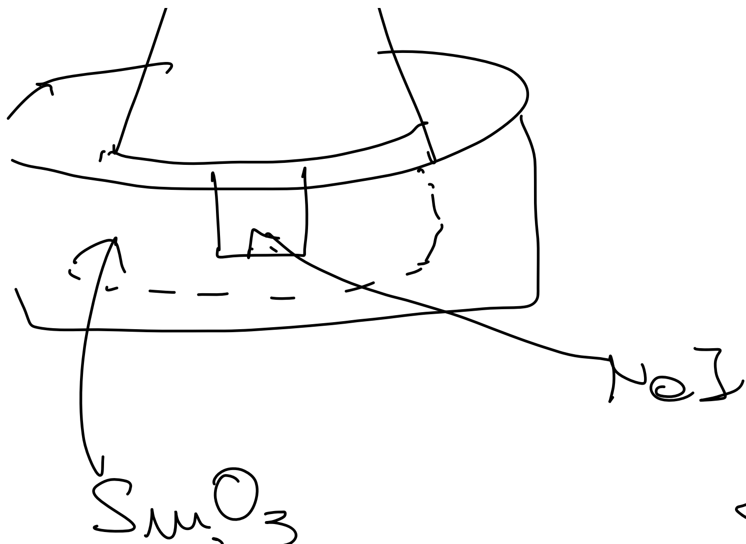
$\downarrow$

$\gamma + Sm \longrightarrow Sm^*$ CAN HAPPEN

$+$ RECALL $h(\gamma) = h(\nu)$!

• How is the exp. setup done?





γ emitted in $-z$ can excitate Sm_2O_3. The disexcitation of the latter gives signal in NaI

$\Rightarrow$ Based on when we see the signal we can know the $h(\gamma)$ and hence $h(\nu)$.

- I focus only on configurations where γ goes in $-z$

  b/c these γ have the right energy and the same h of ν.

- Sm has $J=1$, γ has $J=1$, ν has $J=\frac{1}{2}$
- I try to measure J
-

CASES

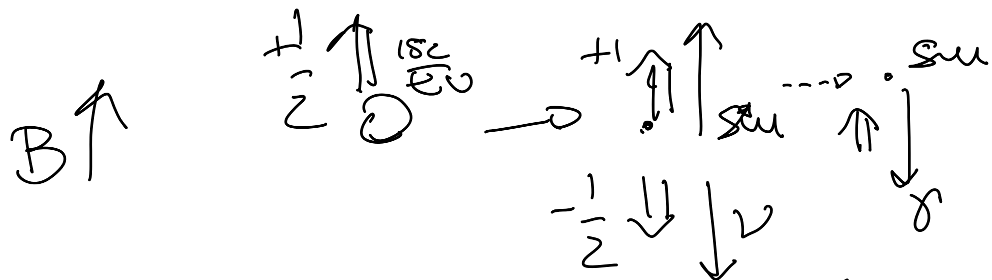
	B↑	B↓
$h(\nu)=1$	NO	YES
$h(\nu)=-1$	YES	NO

counts compatible w/ this config.

← do I see or not signal?

h = +1

B↑

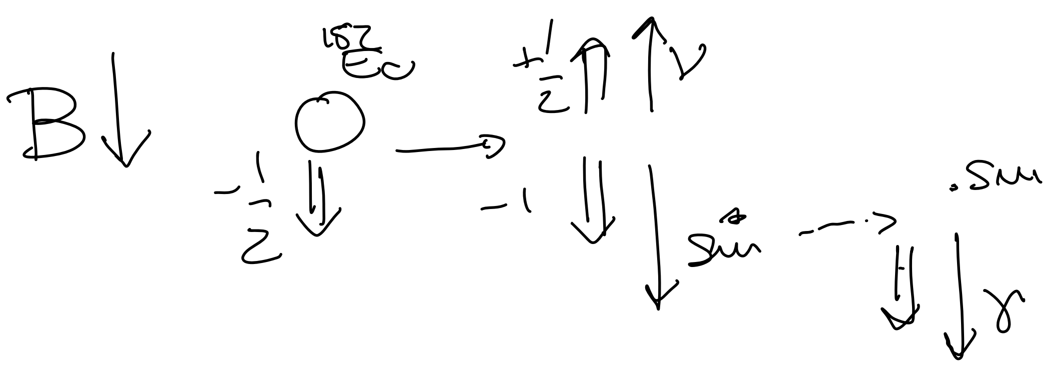


NO BOOST for γ

hence NO SIGNAL

$E_\gamma < \Delta E$

B↓



boost (γ same direction of Sm*)

YES SIGNAL

$\underline{\underline{h = -1}}$

$B \uparrow \quad \tfrac{1}{2} \Uparrow \; {}^{152}\mathrm{Eu}^{0} \longrightarrow -\tfrac{1}{2} \Downarrow \uparrow \nu$

$+1 \Uparrow \downarrow \mathrm{Sm}^{*} \dashrightarrow \Uparrow \downarrow \gamma \quad {}^{\cdot}\mathrm{Sm}$

Boost → YES SIGN.

$B \downarrow \quad {}^{152}\mathrm{Eu}^{0} \quad -\tfrac{1}{2} \Downarrow \longrightarrow -1 \Downarrow \uparrow \mathrm{Sm}^{*} \dashrightarrow \Downarrow \downarrow \gamma \quad {}^{\cdot}\mathrm{Sm}$

$-1 + \tfrac{1}{2} \Uparrow \downarrow \nu$

No boost no sig.

$\Rightarrow \nu$ has $h = -1$

(and $\bar{\nu}$ has $h = +1$)

• Do ν w/ $h = +1$ exist?

$q_\nu = 0$ ⟹ no ELM

$m_\nu = 0$ ⟹ no grav.

$\text{Color}_\nu = 0$ ⟹ no strong

$h(\nu) \neq +1$ ⟹ no weak

⟹ in SM ν w/ $h(\nu) = +1$ do not exist.

<u>However</u> we said that h is not invariant, so it seems also weak interaction fail.

↓

This is b/c what really matters is not helicity but <u>chirality</u>.

Now: helicity is "something we can see", why CHIRALITY tells us "how particles and their antiparticles behave (transform) in weak interaction.

dirac spinor

$$\psi = \left(\frac{1-\gamma^5}{2}\right)\psi + \left(\frac{1+\gamma^5}{2}\right)\psi$$

$$= \psi_L + \psi_R$$

NB $\left[\begin{array}{l}\gamma^5 = i\gamma_0\gamma_1\gamma_2\gamma_3 \\ \gamma^5 = \mathbb{1}\end{array}\right.$

If I call: $\underbrace{\left(\frac{1-\gamma^5}{2}\right)}_{P_L}\psi = \psi_L$

$$\underbrace{\left(\frac{1+\gamma^5}{2}\right)}_{P_R}\psi = \psi_R$$

$$\gamma^5 \psi_L = \frac{\gamma^5 - 1}{2} \psi = - \psi_L$$

$$\gamma^5 \psi_R = \frac{\gamma^5 + 1}{2} \psi = + \psi_R$$

A priori each particle has a L and a R component which transform differently in WEAK INTERACTION.

$\Rightarrow$ In WEAK INTERACTION

particles $\longleftrightarrow$ ψ_L

antiparticles $\longleftrightarrow$ ψ_R

However in principle:

$$\psi_L = a_+ u_+ + a_- u_-$$

For MASSIVE particles both helicity states (u_+, u_-) are possible.

but how a_+, a_- depend on mass?

$$|\langle e_+ | \varphi_L \rangle|^2 \propto \left(\frac{1-\beta}{2}\right)$$

$$|\langle e_- | \varphi_L \rangle|^2 \propto \left(\frac{1+\beta}{2}\right)$$

The positive helicity component of PARTICLES is SUPPRESSED by $1-\beta$

$\Rightarrow \beta \to 1 \quad 1-\beta \to 0$

$\varphi_L \longrightarrow u^-$

For ν where $m \to 0$ $\beta = 1$

$$\psi_L = u^-$$

chirality = helicity

For massive particles a priori no, BUT we need to keep in mind this $(1-\beta)$ suppression factor.

e.g.

$\psi(\ell^-)$ is only $\psi_L(\ell^-)$.

ℓ^- exist w/ $h(\cdot) = +1$ but suppressed by $(1-\beta)$

now $m \nearrow$ $\beta \swarrow$ at fixed p

$m_\mu \sim 100$ MeV $\beta \sim 0.3$

$m_e \sim 0.5$ MeV $\beta \sim 1$

processes w/ righthanded e^- will be suppressed w.r.t same processes w/ μs.

$\pi^- \to \mu^- \bar{\nu}_\mu$

$\pi^- \to e^- \bar{\nu}_e$ this one is favoured by $\rho(E)$

$\pi^- \bullet \dashrightarrow \Uparrow \uparrow e^-$; $\Downarrow \downarrow \bar{\nu}_e$

e^- is produced w/ $h(e^-) = +1$

$\pi^- \to e^- \bar{\nu}_e$ gets a factor $\left(\frac{1-\beta}{2}\right) \sim 0$ of suppression

$$R \; \frac{(\pi \to e\nu)}{(\pi \to \mu\nu)} \sim 10^{-}$$